Do Now · retrieval
Unfamiliar programming challenges

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What is a loop invariant?

Q2

What distinguishes a run from a distinct value?

Q3

Why handle empty arrays explicitly?

Q4

For a=[1,1,2,1], give count of distinct values and runs.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Understand — A run is a maximal group of equal consecutive values, not a unique value across the whole array.

Core idea 2

Apply — Maintain invariant: after index i, the count equals the number of runs in the processed prefix.

Core idea 3

Evaluate — Handle empty input and test all-equal, alternating and separated-repeat arrays.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

run — Consecutive equal-value group.

invariant — Property maintained across iterations.

edge case — Boundary or unusual input.

distinct — Different values irrespective of position.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

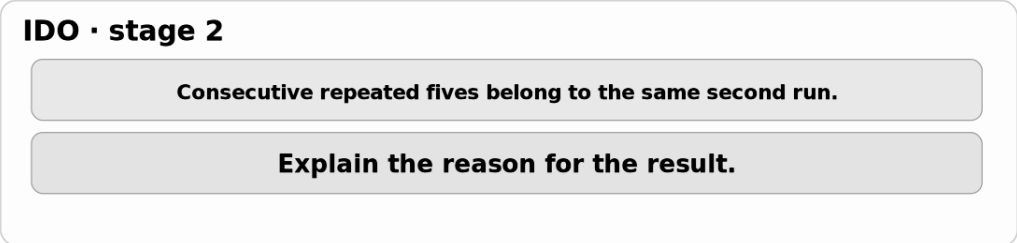
Problem

Trace CountRuns on [4,4,5,5,5].

Follow the worked method

1. Predict first — Consecutive repeated fives belong to the same second run.
2. Explain and check — Explain the reason for the result.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Predict first
There are four runs but only two distinct values.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Trace CountRuns for [3,3,4,4,3,5,5]. State where runs increment. [6]

Question 2

Write a valid C# CountRuns(int[] a) returning runs and handling empty input. [8]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

For [1,2,1], compare runs and distinct counts, and assess time/extra-space complexity of CountRuns. [6]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

Runs always equal number of unique values.

Use key vocabulary from memory

1. Explain run in your own words.

2. Explain invariant in your own words.

3. Explain edge case in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.